

The comparison of ruthenium brachytherapy and simultaneous transpupillary thermotherapy of choroidal melanoma with brachytherapy alone

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ABSTRACT

PURPOSE: To compare the outcomes of combined treatment of choroidal melanoma with ruthenium brachytherapy (BT) simultaneously with transpupillary thermotherapy (TTT) and treatment with BT alone.

METHODS AND MATERIALS: Two matched groups of patients, one treated with BT and simultaneous TTT (Group BT + TTT, $n = 63$), the other treated with BT alone (Group BT, $n = 70$) were analyzed retrospectively. The main outcome measures were rate of tumor regression, recurrences, enucleations, metastases, recurrence-free and overall survival rate, and visual acuity, assessed by Kaplan–Meier analysis.

RESULTS: Patients were matched according to mean age ($p = 0.22$), mean tumor thickness (6.4 vs. 6.25 mm, range 2.5–10.8 mm, $p = 0.59$), and mean length of followup (42 vs. 34.4 months, range 3–109, $p = 0.052$). Tumor largest basal diameter (13.0 vs. 12.9 mm), tumor location, and mean radiation dose (apical 135 vs. 136 Gy and scleral 1294 vs. 1438 Gy) were also similar in both groups ($p > 0.1$). Treatment with BT + TTT resulted in higher rate of tumor regression (63% vs. 49%, respectively, $p = 0.036$), lower 5-year tumor recurrence rate (96% vs. 83%, $p < 0.034$), and higher eye-globe preservation (98% vs. 87%, $p < 0.024$) and recurrence-free survival rates (89% vs. 67%, $p < 0.017$) than treatment with BT alone. There was no difference in complications ($p > 0.5$), metastasis-free (93% vs. 81%, $p > 0.22$) and overall survival rates (91% vs. 81%, $p > 0.39$), or in visual outcomes.

CONCLUSION: Combined treatment of choroidal melanoma with ruthenium BT and simultaneous TTT seems to provide higher local control, eye-globe preservation, and recurrence-free survival rates than treatment with BT alone and results in similar rates of metastases and overall survival. © 2012 American Brachytherapy Society. Published by Elsevier Inc. All rights reserved.

Keywords:

Brachytherapy; Choroidal melanoma; Transpupillary thermotherapy

Introduction

Episcleral plaque radiotherapy is one of the standard treatment options for uveal melanoma. The most common isotopes used for brachytherapy are ruthenium-106 (^{106}Ru) and iodine-125 with their well-known advantages and disadvantages. The most important limitation of ^{106}Ru brachytherapy (BT) is the suitable tumor thickness,

generally considered to be about 5–7 mm (1–4), although there are some studies of ^{106}Ru BT of tumors with thickness up to 11.8 mm (5). The proposal to use BT in combination with a new method of treatment transpupillary thermotherapy (TTT) (6) was associated with the hope, among other things, to make it possible to treat thicker tumors and improve the globe salvage rate (7). By contrast with the dozens of existing studies of TTT as a primary treatment of choroidal melanoma, including those covering the largest groups of treated patients (8–12), only a few reports on the combined treatment of choroidal melanoma with BT and TTT have been published to date (13–18). These studies described nonuniform patterns of application of TTT with BT (just simultaneously (7, 17), simultaneously and with additional TTT during the followup (15), after

Received 7 April 2011; received in revised form 15 August 2011; accepted 8 September 2011.

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